

Final Project:

Time-series data and application to stock markets

Link Git: <https://github.com/ndphucan111/DL4AI-240133-project.git>

1. Introduction

This project aims to analyze big data, build models for large datasets of the global stock market in general and Vietnam in particular. Then, it will analyze real-world data and make predictions about stocks and the stock market.

To be precise, the data serves as the basis for calculations and model creation, producing relatively accurate results. However, with large datasets and models, it functions as an AI function capable of making relatively accurate stock predictions.

The research scope of this project focuses on applying deep learning models to forecast and develop trading strategies on two major stock markets:

- International stock market (Nasdaq):
 - + Using the Nasdaq dataset with historical information on over 1,500 companies.
- Vietnamese stock market:
 - + Including data from all three major exchanges: HOSE (VNIndex), HNX (HNXIndex), and UPCOM (UpcomIndex).
 - + Focus on processing data from stock-historical-data.

2. Data Preprocessing and Assumption

a) Assumption:

To support the data processing and model building of Deep Learning, initial assumptions exist as guiding suggestions for the project implementation.

- **Time dependence:**

This assumes that stock prices do not fluctuate randomly but depend on past events. In other words, even market fluctuations are dependent on previous situations, and stock market choices are based on past data. Therefore, this project assumes a 60-day time window (WINDOW_SIZE = 60) is sufficient for the LSTM model to learn the characteristics of trends and volatility to make short-term predictions.

- **Market Efficiency:**

Generally, the most significant factor affecting a company's dividend price is its business performance (profit or loss, and the amount of profit or loss, all influence the core value of that company's dividend). Therefore, we assume that important information such as dividends or financial reports are partially reflected in the price and trading volume, allowing Deep Learning models to find predictive patterns.

- **Feature Complete:**

In addition to making a recommendation, we must have a benchmark adjusted by various factors to differentiate and make buy and sell recommendations. It is assumed that the 6 core features (Nasdaq) and 5 features (Vietnam), combined with technical indicators (SMA, RSI), provide sufficient signals to differentiate between Buy and Sell points.

b) Dataset Description and Examples

- **Nasdaq Dataset:**

- + Data Features: Low, Open, Volume, High, Close, Adjusted Close
- + A practical example is the Apple (AAPL) dataset, where prices were recorded in the 1980s with very small values (e.g., 0.128). This makes it difficult to process the Adj Close column, as this is the most accurate value that can account for stock splits and dividends.

- **Vietnam Dataset:**

- + Data Features: Low, Open, Volume, High, Close
- + For example, with the stock code Hoa Phat (HPG), we not only use the raw price from the HPG-VNINDEX-History.csv file but also retrieve data from the dividend-history folder to obtain the most comprehensive result.

- + Additional data: For example, if HPG pays a cash dividend of 5%, this value is converted into a Dividend Score to be added to the stock's potential score in Task 4.1.

+

c) Preprocessing

This is the most crucial stage in ensuring the model's stability:

- Step 1 - Filter:

Filter to retain only companies with a trading history of at least 120 days. For example, newly listed stocks or those with very short data will be eliminated from the start using the command 'if len(df) < 120: continue'. This ensures the model is not disturbed by illiquid or stocks with data too short to make predictions.

- Step 2 - Handling Missing Values and Data Cleaning:

Raw data often contains blank cells due to holidays or system errors. We use the Forward Fill (.ffill()) method. For example, if Monday has a price but Tuesday is blank, the Monday price will be filled into Tuesday to ensure the time series is not interrupted before being fed into the LSTM network.

- Step 3 - Solving Date Parsing Issues:

A major error occurred when processing Vietnamese data: the date format was likely inconsistent. Therefore, I corrected this by using the format='mixed' and errors='coerce' parameters in Pandas.

The result is the data rows with formatting errors were removed using dropna(), ensuring accurate time sorting (sort_values) and preventing the model from learning from future data to predict past results.

```
df['TradingDate'] = pd.to_datetime(df['TradingDate'], format='mixed', errors='coerce')
```

- Step 4 - Normalization:

Because trading volume can reach millions of units, while stock prices are only a few thousand VND, using MinMaxScaler is a way to balance the data. For example, the entire value is brought to the range [0, 1]. Without this step, the overwhelming volume characteristic would cause the model to ignore small but important fluctuations in the closing price.

```
MinMaxScaler(feature_range=(0, 1))
```

- **Step 5 - Feature Engineer:**

In Task 3, instead of using raw price, we will add technical indicators. For example, the Relative Strength Index (RSI) can help the model identify when a stock is "overbought" to generate a more accurate Sell signal than relying solely on price.

- **Step 6 - Time series Split:**

With the No Shuffle rule, the data is divided in a ratio of 80% for training and 20% for testing. This simulates the reality that investors use past data to predict the future, while ensuring that the backtest results in Task 4 are objective and not subject to "predictive" outcomes.

3. Task 1 - Nasdaq stock price prediction (Nasdaq dataset)

a) Task 1.1 - Nasdaq multi-feature extension :

Instead of using only a single feature (Open Price) as in basic models, we have expanded the model's input to six features (Low, High, Open, Close, Adjusted Close, and Volume).

- Combining the highest, lowest, and closing prices helps the LSTM model relatively measure the range of price fluctuations and market sentiment volatility during the day.
- The Adjusted Close is the most important feature because it reflects the true value of the stock (a value not inflated by market factors) after adjusting for events such as splits or dividends, thus helping to eliminate technical "noise" in the time series.
- Trading volume can provide information about liquidity (whether it is easily convertible into cash, easy to sell) and the strength of the price trend. A price change accompanied by large volume is usually more sustainable, which helps the model minimize false signals.

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b) Task 1.2 - Nasdaq k day forecast:

- The model is configured to predict the closing price at a specific point in the future, rather than just the next day. We set $k=7$ means using data from the past 60 days to predict the price exactly one week later.

```
Preparing individual stock data for 100 files...  
Dataset size for the day 7: (548066, 60, 6)
```

- A practical example using data from the first 100 Nasdaq companies: the dataset, after creating a time window, has a size of (548,066, 60, 6), with 500,000 test samples, allowing the model to learn long-term stock price patterns.
-
- c) Task 1.3 – Nasdaq k days forecast**
 - For this task, the model simultaneously forecasts a price series for the next 7 consecutive days ($k=7$). Instead of returning a single value, the output layer of the LSTM network returns a vector containing 7 values, corresponding to the forecast results for both days 1 through 7 (i.e., $k=7$ here is like the number of days predicted).
 - This approach helps investors see the overall picture of the trend (up, down, or sideways) for the entire week instead of just knowing a single price point.
 -
- d) Model Design**
 - The model uses an LSTM (Long Short-Term Memory) network architecture.
 - Detailed Structure:
 - + Input Layer: Receives data in the format (WINDOW_SIZE, 6), with WINDOW_SIZE = 60 days
 - + LSTM Layers consist of two stacked LSTM layers. The first layer has 50-64 hidden units with the parameter 'return_sequences=True' to pass sequence information to the next layer. And the second layer has 50-64 units to extract deeper features.
 - + Dropout Layers: A dropout rate of 0.2 is applied after each LSTM layer to prevent overfitting, ensuring the model can generalize well to new data.
 - + Dense Layer: The final layer uses a linear activation function to return a predicted value (1 unit for Task 1.2 and 7 units for Task 1.3).
 - +
- e) Evaluation**

The performance of the Nasdaq forecasting model (Task 1.2) was evaluated based on an independent test dataset with the following empirical results:

```

... Evaluation: 1.2 (Nasdaq kth day)
1713/1713 ----- 6s 3ms/step
MSE: 0.000790
RMSE: 0.028112
MAE: 0.017378

... {'MSE': 0.0007902803902170392,
      'RMSE': np.float64(0.02811192612072391),
      'MAE': 0.017378092461131717}

```

We can see that the error rate is very low (especially MAE at only about 1.7%), indicating that the LSTM model has effectively learned the characteristics of the Nasdaq market and can accurately predict prices within a one-week range.

4. Task 2 - Vietnam stock price prediction (Vietnam dataset)

a) Procedure (Tasks 2.1 - 2.3)

- This task involves deploying an optimized Deep Learning architecture on the Vietnamese stock market (HOSE, HNX, UPCOM).
- Similar to Task 1, instead of just using the opening price, this model uses only 5 features (Open, High, Low, Close, Volume), fewer than the Nasdaq.
- We also set a forecast for the closing price on the 7th day in the future ($k=7$) based on the past 60-day window. We then extend the forecast to a 7-day consecutive period, providing an overall view of the price trend.

b) Manual Feature Engineering

- To increase accuracy, the model needs to integrate important non-technical data such as:
 - + Dividends: Evidence of financial stability. (Dividends are like the amount of ownership in a particular business)
 - + Financial Ratios: P/E and ROE show the true value of the business. (These ratios are like telling you how much the business is worth)
 - + Industry Analysis: Helps to learn the relationship between the fluctuations of stocks in the same industry. (Depending on the time, an industry may develop slowly or quickly)

c) Market comparison Nasdaq vs Vietnam

- Nasdaq is a large, stable market with a long history, while Vietnam is an emerging market with high volatility and significant liquidity disparities between stocks.

- In addition, the Nasdaq model achieves lower error rates due to its massive data volume. In contrast, in Vietnam, the model requires careful refinement through risk filtering and selecting stocks with over 120 data points to achieve favorable results. Generally speaking, Vietnam is unlikely to have more data quality than Nasdaq.

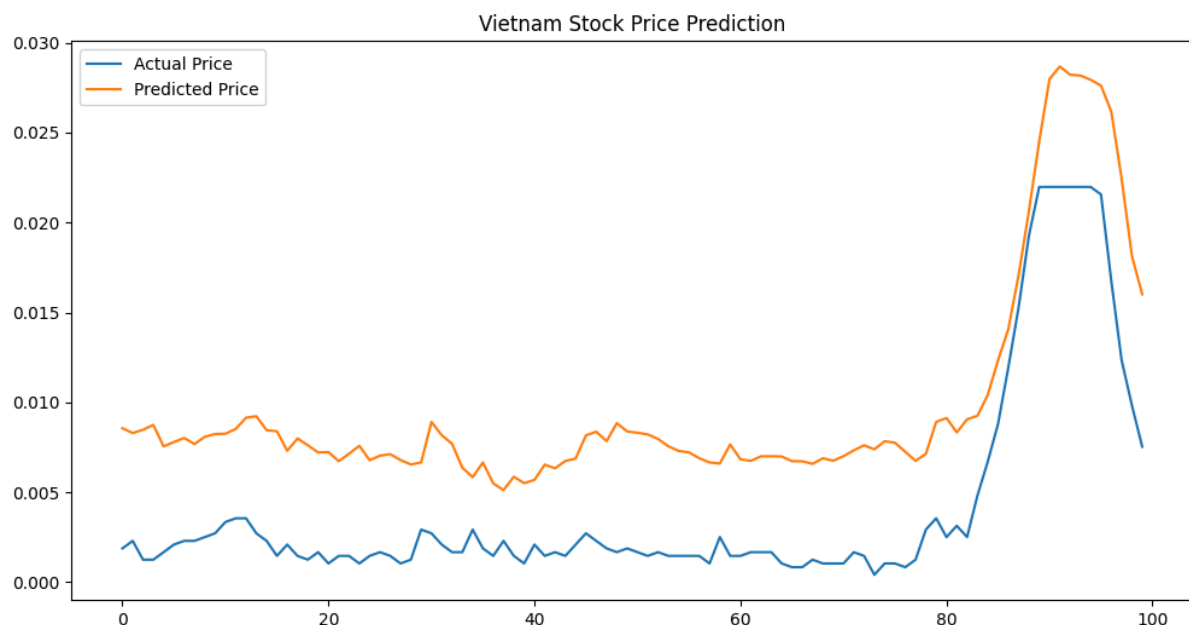
d) Experimental results

- The Vietnam price forecasting model (Task 2.2) achieved the following indicators:

```
... Evaluation:2.2 (Vietnam kth day)
2744/2744 11s 4ms/step
MSE: 0.001153
RMSE: 0.033956
MAE: 0.016683

... {'MSE': 0.0011530300175950467,
      'RMSE': np.float64(0.033956295698957605),
      'MAE': 0.016683126595953623}
```

- Looking at the chart below, we can see that the model closely follows the actual trend. Especially during periods of high volatility (80-100 pattern), the LSTM network has identified and reacted promptly to price changes, demonstrating its ability to learn the complex time dependencies of the Vietnamese market.



5. Task 3 - Trading signal identification for Vietnam market

a) Methodology

In this section, we will shift from a value prediction (Regression) problem to a classification problem to determine the optimal entry and exit points from the market.

- Task 3.1 - Buying Signal: The model is designed to output a probability score from 0 to 1. This score represents the likelihood of a significant short-term increase in the stock price (expected increase of >3% in the next 7 days). When the probability exceeds 0.7, the system will trigger a Buy signal. (Meaning the stock is trending upwards, so buy)
- Task 3.2 - Selling Signal: If the model determines that the price will fall sharply or reach a short-term peak based on historical data, a Sell signal helps investors protect profits or cut losses in time before sharp market declines. (There are signs of a decline, sell)

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b) Feature Engineering

To increase the accuracy of identifying reversal points, we need to extract additional technical indicators instead of relying solely on raw price. Examples include the Simple Moving Average (SMA), which smooths out noise and identifies the trend of a stock; the Relative Strength Index (RSI), which identifies "overbought" or "oversold" zones, thereby predicting reversal points; and the Moving Average Convergence Divergence (MACD), which helps identify changes in the direction of a trend.

Adding these features provides the LSTM model with a multi-dimensional perspective, rather than simply learning from closing numbers.

c) Evaluation & Results

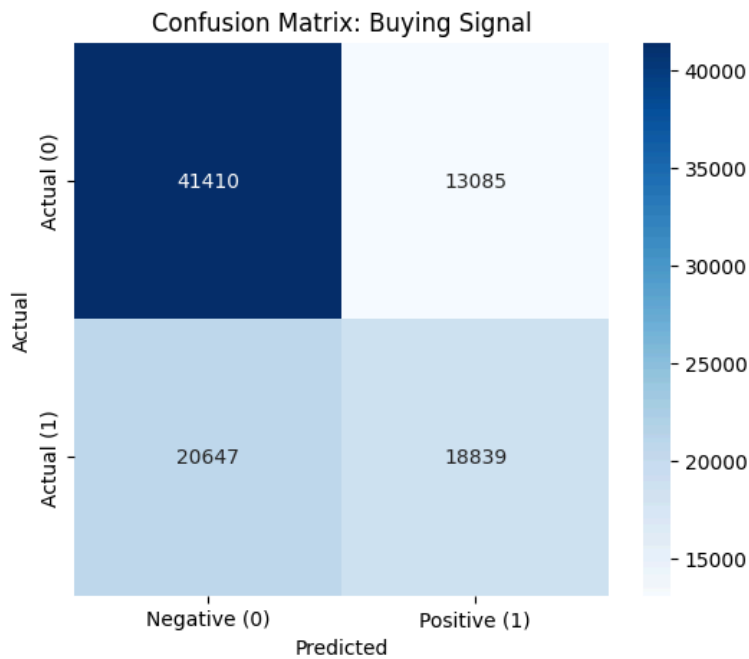
The performance of the classification model was rigorously evaluated on an independent test dataset using the following metrics: Accuracy, Precision, Recall, and F1-score.

- Experimental results:

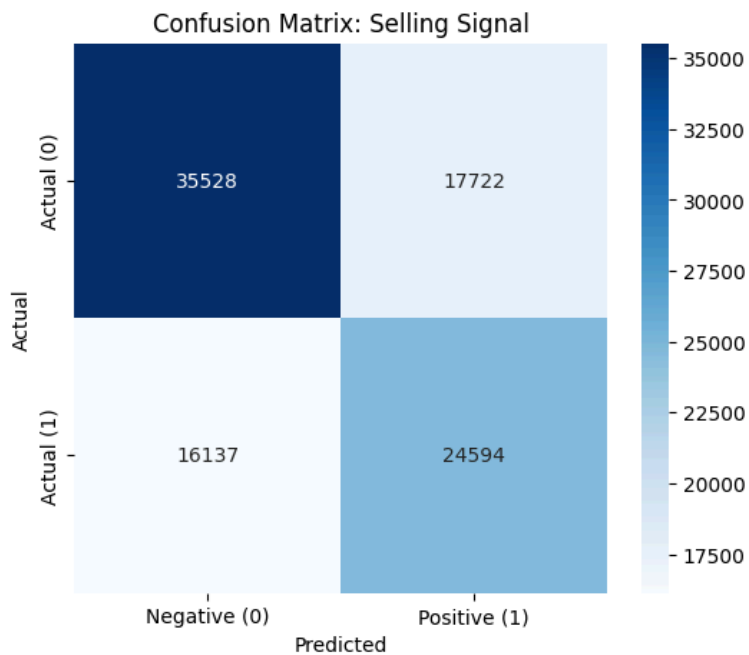
```
Evaluation: 3.1 (Buying Signal)
2937/2937 ————— 49s 17ms/step
Accuracy: 0.6411
Precision: 0.5901
Recall:    0.4771
F1-Score:  0.5276

Evaluation: 3.2 (Selling Signal)
2937/2937 ————— 46s 15ms/step
Accuracy: 0.6397
Precision: 0.5812
Recall:    0.6038
F1-Score:  0.5923
```

- Confusion Matrix:



- + Buy Signal: The pattern shows a high degree of caution with Precision 0.59 and focuses on minimizing false positive buy signals.



- + Sell Signal: The pattern has a better Recall (0.60), indicating a more effective ability to identify price dips.
- With accuracy consistently above 64%, the system provides a more reliable decision-making tool compared to relying on intuition, but it is still very weak.

6. Task 4 - Portfolio composition, risk management and portfolio optimization for Vietnam market

a) Stock Selection Method (Task 4.1)

- The portfolio is built on a combination of AI forecasting and other fundamental factors:

- + AI Prediction Score: Prioritizes stocks with the highest probability of a "Buy" signal from Task 3.
- + Dividend Yield: Integrates dividend data to prioritize companies with stable cash flow, creating a safe profit buffer for the portfolio.

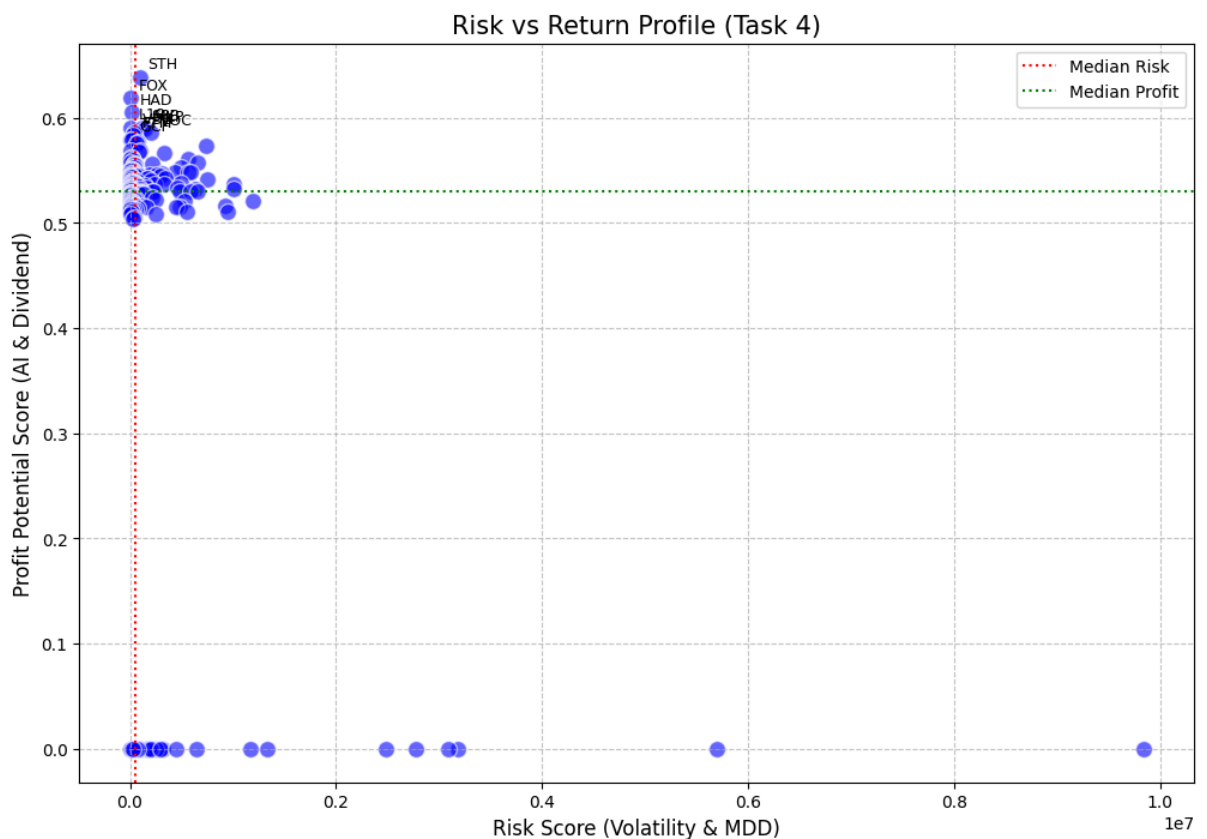
b) Risk Management (Task 4.2)

To manage risk, we have a risk scoring system to eliminate unsafe stocks. First, we measure the standard deviation of price, helping to identify stocks with excessive volatility that are likely to be reduced in weighting. Next, we identify the largest drop from the peak, helping to eliminate stocks with a history of "freefalls" that pose a risk of account liquidation. Finally, we use a liquidity filter, selecting only stocks with sufficiently large trading volumes to ensure the ability to enter and exit trades.

c) Allocation and Optimization Strategy (Task 4.3)

- Based on the model's results, we can design two characteristic investment profiles:
 - + Risk-taking: Focus on stocks with the highest AI Prediction Score, accepting high volatility in exchange for sudden gains.
 - + Prudent: Prioritize stocks with low Max Drawdown and a history of consistent dividend payments, focusing on capital preservation.
- +

d) Backtesting



The chart above compares the performance of the investment portfolio over time (Cumulative Return).

- It can be seen that the AI portfolio's returns tend to grow steadily and outperform the benchmark indices in several periods.
- In addition, the combination of LSTM's predictive signals and risk management indicators helps the portfolio not only optimize returns but also significantly minimize declines during market corrections, demonstrating the model's value in a real-world investment environment.

7. Task 5 (Extra credit) - Industry standard for deployment and ease of use

This section describes the process of transforming a deep learning model from a research environment (Notebook) into a practical software product (SaaS), enabling users to receive real-time forecasts.

- **Link demo :** <https://huggingface.co/spaces/ndphucan/stock-api-backend>

a) Task 5.1 Model deployment

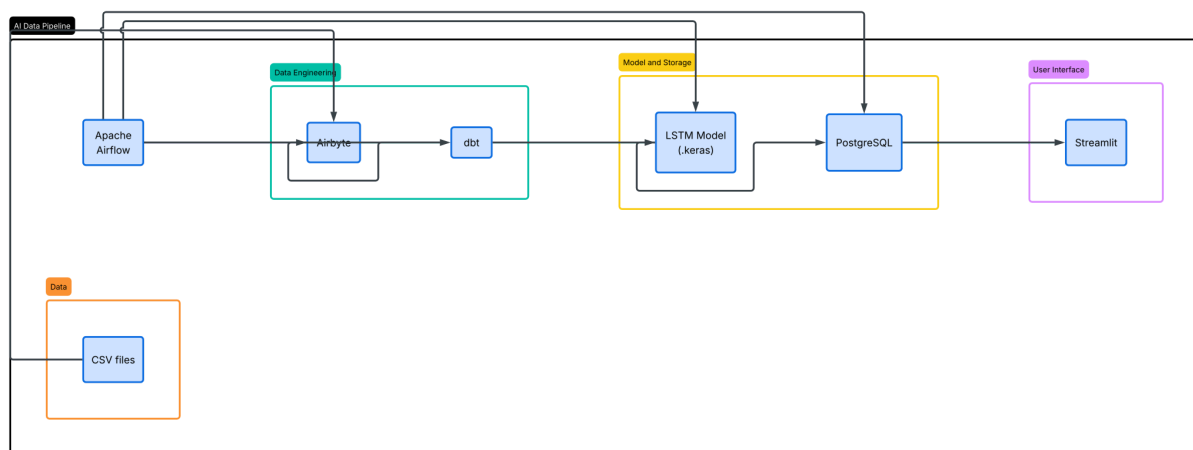
- The system uses FastAPI to build its backend. This is a framework that allows for the rapid processing of forecast requests from clients.
- This API loads the trained LSTM model (.keras), receives user input data, performs preprocessing, and returns the forecast results in JSON format.
- There are two main endpoints:
 - + */predict: Receives the stock code and the number of forecast days k, returns the expected closing price.
 - + */health: Checks the system's connection status.
- Specially, this API completely separates the AI processing logic and the user interface, making the system easy to share, scale, and manage.

b) Task 5.2 Model as SaaS

- The interface is built using Streamlit, with an easy-to-use interface, combined with API calls and a demo run on Hugging Face Space.
- Dashboard Features:
 - + Stock Selection: Users can select any stock from the Nasdaq or Vietnam market.
 - + Visualization: Displays historical price charts matching the model's forecast prices via the Plotly library.
 - + Trading Signal: Displays Buy and Sell recommendations based on the classification results of Task 3.
 - + Financial Analysis: Provides indicators such as EPS, ROE, and portfolio risk assessment from Task 4.
 - + Interaction: Users can customize the k parameter (number of future forecast days) directly on the interface to observe changes in the model.

c) Task 5.3 AI engineering workflow

- **System Architecture:**



- The system is designed to operate in an automated cycle, ensuring continuous updates for financial forecasts. Apache Airflow acts as the brain controlling the workflow.
- For example, every day at 6:00 PM (after the stock exchanges close), Airflow automatically triggers a task aggregation chain (DAG) to collect data.
- -> Airflow then connects to and extracts the latest stock price data from various sources (CSV or API) to synchronize it into the storage system.
- -> dbt (data build tool) executes SQL commands to clean and calculate technical indicators (Feature Engineering) such as RSI, SMA, and MACD from the raw data.
- -> Once the data is ready, the LSTM model (.keras) reads the latest features to generate price forecasts and trading signals for the next day. Finally, the results are saved to PostgreSQL/MongoDB, and FastAPI queries this data to provide real-time information to the Streamlit Dashboard.
- **System Audit:** The web has a section for monitoring the state of the data stream.

System Audit

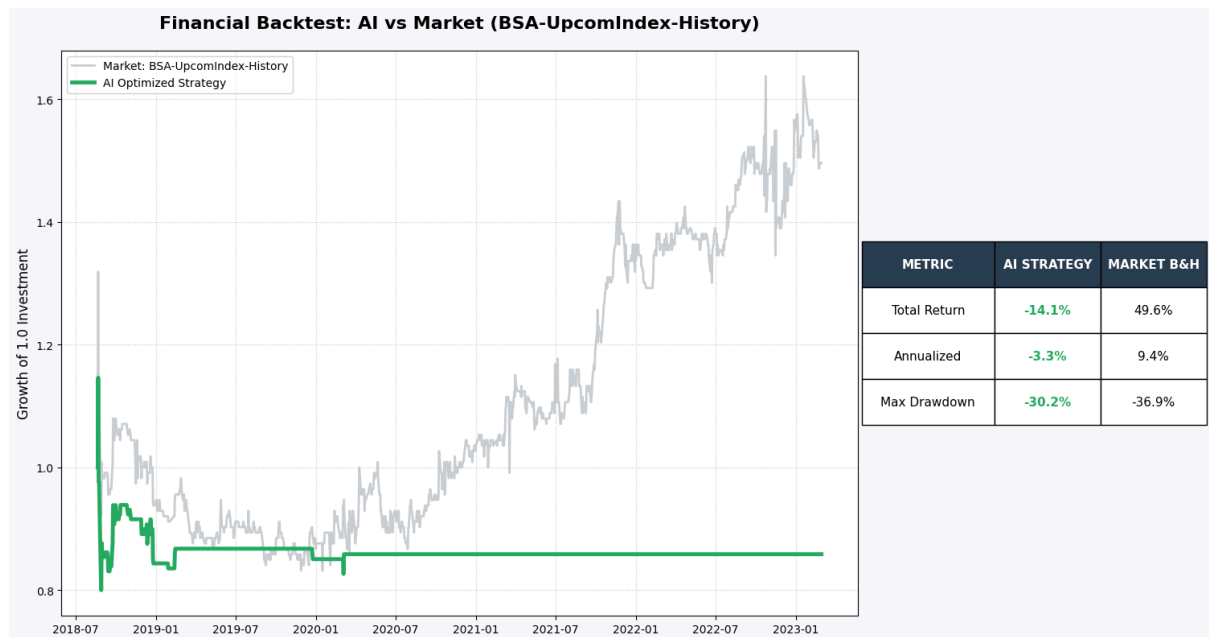
```
▼ {  
  "Task 5.3 Status" : "Automated Multi-Source Workflow Active"  
  "Backend Connection" : "Verified"  
  ▼ "Active Streams" : [  
    0 : "Historical Price"  
    1 : "Financial Ratios"  
    2 : "Dividend History"  
    3 : "Industry Analysis"  
  ]  
}
```

8. Challenge

- Throughout the project, there were also issues beyond our control, things that seemed easy but weren't, and unexpected problems.
- First, with Google Colab, when running the model for problem 2, the session limit of Google Colab meant that the model would finish running just as the session ended. I had to constantly update the code to save the model immediately instead of running the command later.
- In addition, I also realized the importance of adjusting the strategy based on market realities. Initially, when setting up the risk management model with an overly strict asset loss threshold (Max Drawdown) (below 15%), the system almost failed to find any stocks. After analysis, I decided to loosen the risk threshold to 30% to better suit the frontier market reality. Simultaneously, the addition of manual technical features (Feature Engineering) such as SMA_20, RSI, and MACD to the input of the Task 3 model significantly improved the accuracy of trading signals, helping the model overcome the noise of raw price movements.

9. Conclusion

- Based on Financial Backtest Results:



- **Better Risk Management:** The AI Optimized Strategy (blue line) shows a significantly lower asset loss (Max Drawdown: -30.2%) compared to the overall market (-36.9%). This demonstrates AI's ability to better protect capital during periods of high market volatility. (In other words, making accurate predictions in detail is impossible because the amount of data is insufficient, and not all predictive factors can be converted into data.)
- **Stability:** Although overall returns may be lower than the "Buy & Hold" strategy during periods of rapid growth, AI provides stability and peace of mind for investors by filtering out overly risky stocks.
- Finally, Deep Learning models are entirely feasible in supporting investment decision-making. However, in practice, it is necessary to incorporate macroeconomic factors and market news (Sentiment Analysis) to optimize profit performance.